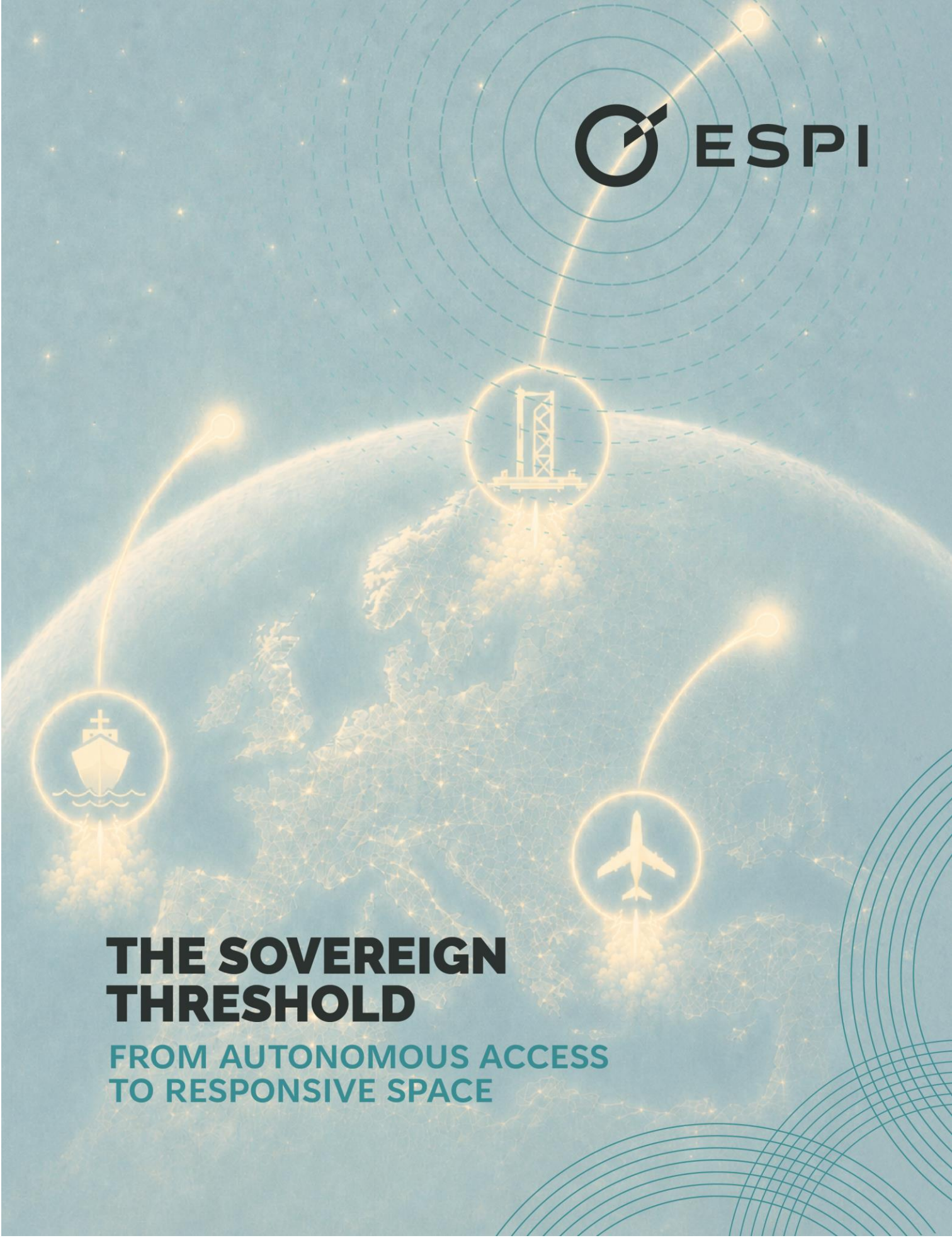


THE SOVEREIGN THRESHOLD

FROM AUTONOMOUS ACCESS TO RESPONSIVE SPACE

The Sovereign Threshold: From Autonomous Access to Responsive Space

Amid a deteriorating geopolitical environment, increased reliance on space infrastructure by modern society, and increasingly confrontational behaviours in space, the **concept of responsive space capabilities** is growing significantly in relevance.

The trend is underpinned by a growing number of agreements explicitly seeking to diversify launch infrastructure (spaceports, launch vehicles, logistics), aiming to increase resilience and responsiveness. Moreover, **a higher launch cadence** (*Figure 1*) creates conditions where responsive launch is becoming a feasible concept even without dedicated infrastructure.

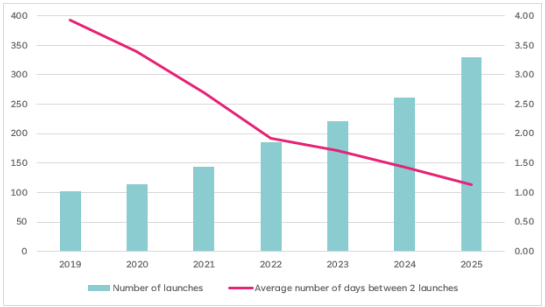


Figure 1: Global launch cadence 2019-2025 (credit: ESPI)

While definitions of responsive space differ, they generally include **replacing a lost asset/service or adding an asset/service to enhance resilience and redundancy** against threats and risks. Broadly, responsive space encompasses rapid launch, data sharing, cooperation, or in-orbit operations. Drivers behind responsive space can range from intended attacks through unintended damage caused by space debris collisions or space weather, to new operational needs (e.g. increased revisit rates).[1]

> "We must greatly expand our ability to keep pace with the threat. [...] We need commercial mission partners to build the capabilities to replenish our military space assets. [...] I really like the tactically responsive space concept [...] including rapid preparation of rockets and payloads and conducting immediate launch turnarounds, and we are seeing that in the commercial industry right now."
> **- Gen. Dickinson, Head of the U.S. Space Command (November 2022)**

This policy brief primarily addresses the **launch element** of responsive space.

1 GLOBAL DEVELOPMENTS IN RESPONSIVE SPACE

While Europe has been disentangling itself from the launcher crisis, both the U.S. and China are making strides towards operational responsive space systems by diversifying their portfolio of spaceports and by developing new launch systems.

1.1 Responsive Space Developments in the United States

Looking at the evolution of responsive space in the U.S., the Department of Defense launched the "Operational Responsive Space (ORS)" initiative and a dedicated office i**n 2007**, supported by a three-tiered definition: (1) rapid exploitation of existing capabilities; (2) use of existing technologies and capabilities to replenish, augment and reconstitute;

[1] DLR, Responsive Space Cluster Competence Centre (RSC³): Responsive Space (link); European Space Policy Institute, ESPI Report 22 (2010): Responsive Space for Europe –(link); Wolfgang Jung and Lt Col. Tim Vasen (2021): Responsive Space for NATO Operations, JAPCC, Journal 31, February 2021 (link)

and (3) development of new technologies for the same purposes.[2] The first U.S. attempt to demonstrate responsive launch was the technology demonstrator "Trailblazer", which failed on a Falcon 1 launch in 2008. After several missions (TacSat, ORS-1,-2,-3), the ORS Office was transformed into the Space Rapid Capabilities Office (SpRCO) in 2018, and the U.S. Tactically Responsive Launch initiative was launched in 2021, with TacRL-2.

Mission	Actors	Objectives & Characteristics
TacRL-1 Trailblazer (2008)	**Mission:** SpaceDev **Launcher:** SpaceX' Falcon-1	Technology demonstration mission, with the objective to demonstrate: • Demonstration of a flexible, modular commercial bus design using off-the-shelf components • Trailblazer selected a few weeks before launch
TacRL-2 Odyssey (2021)	**Launcher:** Northrop Grumman's Pegasus XL	SDA technology demonstration, with the objective to demonstrate: • Satellite manufacturing in less than 1 year • Assembly integration and testing within 4 months • 21-day-notice to get ready for flight (horizontal air launch)
TacRS-3 Victus Nox (2024)	**Mission:** Millennium Space **Launcher:** Firefly's Alpha Rocket FLTA003	SDA/RPO mission, with the objective to demonstrate: • Satellite development in under 8 months • Launch order, integration, and launch readiness in under 58 hours • Rapid on-demand launch in under 24 hours • Deployment and in-orbit operations in 37 hours after launch
TacRS-4 Victus Haze (moved to 2026, launched in June 2026)	**Mission:** Rocket Lab **Launcher:** Rocket Lab Electron **Mission:** True Anomaly **Launcher:** SpaceX's Falcon 9	SDA/RPO mission, with the objective to demonstrate: • more complexity than Victus Nox, multi-launch, multi-vehicle mission with 2 launchers & 2 payloads • an end-to-end capability to rapidly respond to threats • entering into Hot Standby Phase before notice to launch • launch in 24h within first available window (done in 16,5 hours) • after reaching orbit, spacecraft are rapidly commissioned to conduct dynamic space operations/ responsive space scenarios
Victus Sol (late 2026)	**Mission:** TBA **Launcher:** Firefly's Alpha	SDA/RPO mission, with the objective to: • further build upon the capabilities demonstrated by Victus Nox • focus on fast payload delivery
Victus Salo (late 2026)	**Mission:** Impulse Space **Launcher:** SpaceX's Falcon 9	SDA/RPO mission, with the objective to demonstrate: • rapid deployment and on-orbit manoeuvring to counter threats to spacecraft, Mira vehicle to operate in LEO • how prepositioned capabilities can improve responsiveness to in-orbit situations, enabling SDA's dynamic operations aspect
Victus Surgo (2026/27)	**Mission:** Impulse Space **Launcher:** SpaceX's Falcon 9	Rapid in-orbit manoeuvring, high-energy transfers to GTO: • responsive space delivery of Helios rideshare payload to GEO, transport Mira vehicle from LEO to GTO on a Helios kick stage • Helios and its payloads will launch on a SpaceX Falcon 9 rocket

rapid launch
rapid deployment & activation
rapid procurement and payload integration
rapid in-orbit operations
Beyond LEO

Figure 2: Evolution of US Tactically Responsive Launch - Space Concept (credit: ESPI)

The U.S. approach is steadily transitioning, from a responsive **launch** focus (Tac**L**) to a broader **Responsive Space** focus (TacR**S**) concept. This includes both rapid deployment and activation, procurement, payload integration, as well as an in-orbit operations.[3] The U.S. is further expanding its responsive space portfolio and recognises not only the operational value of responsive space, but also the deterrence value.[4]

[2] European Space Policy Institute, ESPI Report 22 (2010): Responsive Space for Europe (link).
[3] ESPI Yearbook 2023 (2024): Responsive Space, Chapter Policy & Programmes (link)
[4] Aviation Week (2025): The Pentagon Is Pushing The Limits Of Responsive Space Missions (link)

1.2 Responsive Space Developments in China

Moving east, China has showcased major advances in responsive space since 2022.[5] In June 2022, China launched two rockets within 24 hours from two different spaceports, and in September 2022, China performed two launches within just two hours from two spaceports.[6] When comparing U.S. and Chinese approaches, it is noteworthy that they - at least for now - differ significantly. While the U.S. approach is based on private launch service providers and reusable launch vehicles, China's approach is based on rapidly deployable and smaller solid-fuel launchers. A study by Georgetown's Center for Security and Emerging Technology states that China's rapid increase in launch cadence - as well as the diversification of its mobile small solid-fuel launchers - has put China ahead of the U.S. when focusing solely on responsive launch.[7]



Figure 3: Launch sites in China (Credit: CNES)



Figure 4: Number of Launches U.S. vs China (Credit: ESPI)

The United States, however, still leads the way in terms of pure launch cadence, theoretically providing more opportunities for high-priority payloads to be integrated on "*scheduled*" rather than "*chartered*" services. Moreover, the United States, supported by its extensive and vibrant industrial base, seems to be the frontrunner in broader responsive space frameworks, including rapid procurement and in-orbit operations.

2 INTERNATIONAL PARTNERSHIPS FOR (RESPONSIVE) LAUNCH

Beyond domestic access to space infrastructures, both the U.S. and China follow a strategy of diversifying launch options through bilateral partnerships and investments in a bid to improve their resilience and responsiveness.[8]

The U.S. has been concluding bilateral partnerships to diversify launch territory options, including for responsive space.

This is reflected in **Technology Safeguard Agreements (TSA)** signed with **Australia, New Zealand, Japan, Brazil,** the **UK, Norway** and **Sweden**.

Technology Safeguard Agreements protect sensitive U.S. technology by providing the legal and technical framework and setting the standard for the proper handling of sensitive technology by other states in this partnership for U.S. satellite launches (and return) from foreign spaceports.

[5] ESPI Yearbook 2022 (2023): Responsive Space (link); Map Visual: CNES (2026): ChinEspace, L'espace en Chine, Février 2026.
[6] Space.com (2022): China launches 4 satellites with 2 rockets in 2 days, June 2022 (link); SpaceNews (2022): China performs two launches inside two hours, September 2022 (link).
[7] Sam Bresnick (2023): China's rapid space launch advantage, and how the U.S. can try to counter it, Breaking Defence, (Link); Corey Crowell Sam Bresnick (2023): Defending the Ultimate High Ground: China's Progress toward Space Resilience and Responsive Launch. GeorgeTown University (link); Wei Long (2000): China to Develop Solid Propellant Rocket, Space Daily (link)
[8] ESPI Director's Perspective (2023): From Djibouti and the Northern Territories to the use of space – and outside perspective (link).

The TSA between the U.S. and Japan opens the door to U.S. launch operations from Japanese soil, including responsive launch. Independently, Japan is building own capabilities: its first SDA Guidelines (July 2025) commit to responsive space systems, already reflected in a MoD contract with Astroscale for a Responsive Space System Demonstration Satellite Prototype.

In practice, U.S. cooperation with European countries (primarily Sweden, Norway, the UK) covering responsive launch takes form through bilateral agreements as well as in the frame of NATO STARLIFT. Following the signed TSAs, commercial pathways have followed, whereby U.S. launch companies set up agreements with Swedish, Norwegian, and UK launch site operators:

Norway	Sweden	United Kingdom
TSA signed in January 2025	TSA signed in June 2025	TSA signed in June 2020
Agreement between Andøya spaceport and Astrobotic signed in August 2025.	Agreement between Firefly Aerospace and Swedish Space Corporation signed in June 2025.	Lockheed Martin established a relationship with SaxaVord to launch a satellite for the UK government.

Similarly to the U.S., China is expanding its launch options through international partnerships. In 2023, Djibouti unveiled a $1 billion programme for an orbital spaceport in cooperation with two Chinese companies, Hong Kong Aerospace Technology and Touchroad International Holdings Group. However, public sources suggest the project is on hold due to regulatory hurdles and might not proceed. More recently, an LoI was signed with Malaysia in April 2025 for a feasibility study regarding Pahang International Spaceport, which could become China's first overseas equatorial launch site. If approved, the project is estimated to be completed in the next 3-5 years.[9]

2.1 Multilateral cooperation in responsive space

Multilateral cooperation in (or including) responsive space includes the dedicated NATO STARLIFT project and the RSC MoU. Moreover, other international multilateral initiatives, e.g., the CSpO initiative, include a responsive space dimension.

NATO STARLIFT	Launched in 2024 to develop a resilient, responsive and cost-effective network of capabilities to launch assets at short notice from spaceports across NATO, supported by Belgium, Finland, France, Germany, Hungary, Italy, Luxembourg, the Netherlands, Norway, Spain, Sweden, Türkiye, the UK and the U.S. Canada is expected to become a member. Beyond allies, NATO is in talks with South Korea, Japan, Australia and New Zealand.
Responsive Space Capabilities MoU	Multilateral agreement from 2014, facilitating R&D cooperation, information exchange, and harmonising military requirements for rapid, cost-effective space access, includes Australia, Canada, Germany, Italy, Netherlands, New Zealand, Norway, UK, US, Spain, Sweden. Denmark and Poland are in talks to join.
Combined Space Operations (CSpO) Initiative	Multilateral partnership to improve cooperation, coordination, and interoperability in national security space activities. One of its three objectives is "Space Mission Assurance", emphasising the establishment and maintenance of a robust, responsive, and interoperable space infrastructure. Members are Australia, Canada, France, Germany, Italy, Japan, New Zealand, Norway, the UK, and the U.S.

[9] Marion Douet (2023): Djibouti announces construction of first spaceport in Africa. Le Monde (link); Andrew Jones (2025): China and Malaysia to study international equatorial spaceport project. SpaceNews (link); Lauren Ploch Blanchard (2025): China's Engagement in Djibouti. Library of Congress (link)

3 RESPONSIVE SPACE IN EUROPE

While the U.S. introduced the concept of responsive space in 2007, ESPI drew recommendations for Europe based on the analysis of the U.S. approach in 2010. European ambitions in responsive space have received a boost from both the European launcher crisis and the rising military launch demand, especially since the start of the war in Ukraine. ESPI was also involved in the EDF Research Project REACTS (Responsive European Architecture for Space) from 2023 – 2025, providing policy, governance and legal/regulatory perspectives.

In Europe, responsive space, while not a novelty, remains nascent in terms of technological maturity, and even more so in terms of institutional and operational readiness.[10] To date, it primarily focuses on the responsive launch element.[11] On the hardware side, launchers developed under ESA's European Launcher Challenge, while not purpose-built for responsive space, are set to play a role in future European systems. The EDF project REACTS focused on providing a concept for a European Responsive Space System and Governance.[12] Moreover, a set of launch sites across the European continent is being established, with sites in Northern Europe (Sweden, Norway) and the UK expected to become operational, while several other sites are being planned or under consideration (Portugal, Spain, Italy, Germany, Denmark).

"Until 2030, we plan to finance projects worth €35 billion. [...] **Germany also needs secure, on-demand transport capacities into space**. Our focus is on a mix of small carrier rockets enabling flexible deployment and, in the medium term, European heavy-lift launchers to be developed"
-B. Pistorius, Germany's Minister of Defence (September 2025)

At Member State level, **Germany, France, Italy, the UK, the Netherlands, Luxembourg and Norway** have all highlighted responsive space in their **national strategies**. Germany founded the DLR's Responsive Space Centre (RSC³) in 2020 and highlights the need for responsive space in its security strategy (2023) and its national space safety and security strategy (2025). Beyond policy, Germany has enabled the emergence of three launch companies, supported reintroducing concepts such as the Polaris spaceplane, and is continuing plans for a North Sea offshore launch platform. Two weeks before Isar Aerospace's second launch attempt from Andøya in March 2026, a high-level visit of German Chancellor Merz, Defence Minister Pistorius and Space Minister Bär in Norway resulted in the launch of a **German-Norwegian space working group** in areas of critical importance to European security and preparedness, including launch capacity at Andøya Spaceport.[13] France also continues to develop responsive launch capabilities, announcing additional €4.2B funding for military space activities through 2030, including for reusable and low-cost launcher development and ground and orbital response capabilities. The **2025 French Space Strategy** positions **responsiveness** as a core feature it strives to develop within its ecosystem.

"Faced with this threat, France is acting with a clear strategy: resilience, responsiveness, and the ability to act in and toward space"
– E. Macron, President of France (November 2025)

[10] European Space Policy Institute, ESPI Report 22 (2010): Responsive Space for Europe (link).
[11] ESPI participated in the recent EDF-funded REACTS project. Subsequently, awards for three Mobile Responsive Launch System Studies are being funded by DG DEFIS.
[12] European Commission EDF Factsheet (2023): REACTS Responsive European Architecture for Space (link).
[13] Business Norway (2023): Norway zooms towards first on-demand responsive space launch. February 2023 (link)

On the back of publicly-driven efforts, industry-driven dynamics are taking shape, notably agreements between launch providers and spaceports in Europe for access to space and for responsive launch. A LoI signed between **Rheinmetall** and **Andøya Space** for flexible, rapid response and deployment services, including Tactical Responsive Launch capabilities was signed in September 2025.[14] In May 2026, an Italian Consortium completed a suborbital demonstration of an **air-launched rocket system** - a step towards diversification.[15]

In a similar fashion to U.S. launch operators partnering with European spaceports under the TSA framework, PLD Space (Spain) and Hyimpulse (Germany) entered into agreements with **Oman's Etlaq Spaceport**. Hyimpulse has also already launched suborbitally from **Australia's Koonibba Test Range** and signed an agreement with Southern Launch to explore the possibility of **launching its SL1 rocket from Australia's Whalers Way Orbital Launch Complex**. Isar Aerospace (Germany) is partnering with Maritime Launch Services to launch from **Canada's Nova Scotia**, while plans of Italy refurbishing the **Kenya-based offshore launch facility** have re-emerged on several occasions.[16] Moreover, NATO Starlift is opening for cooperation beyond allies, with international partners, eg RoK, Japan, Australia and New Zealand.[17]

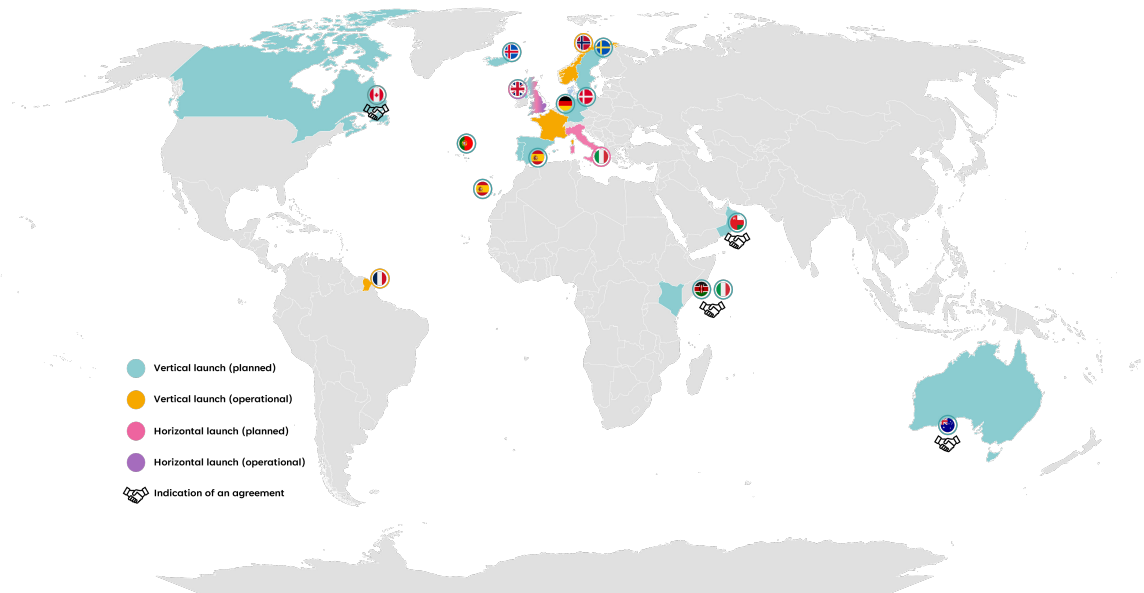


Figure 5: Global Map of planned and operational European launch sites and launch agreements (credit: ESPI)

[14] Andøya Space, Andøya Space and Rheinmetall Nordic sign landmark LOI to enhance European space capabilities, 1 Sep 2025 (link).
[15] Andrew Parsonson: Italy Completes Air-Launched Rocket Demonstrator Test. European Spaceflight, 11 May 2026 (link).
[16] PLD Space: PLD Space expands its global launch capacity with its second spaceport, Etlaq in Oman. PLD Space, 20.02.2025, (link); HyImpulse: HyImpulse to Explore Launch Operations in Oman Under New LOI with Etlaq Spaceport, 2.6.2026 (link); Andrew Parsonson (2024): HyImpulse Signs Agreement to Launch SL1 Rocket From Australia, European Spaceflight, June 2024 (link); Andrew Parsonson (2024): Italy to Reopen Kenya-Based Offshore Launch Facility. European Spaceflight, 25.11.2024 (link). Isar Aerospace (2026): Isar Aerospace partners with Maritime Launch Services for orbital launch readiness from Nova Scotia, 26 May 2026 (link).
[17] The Asia Business Daily (2026): NATO Proposes "StarLift" Space Security Cooperation to Four Countries Including South Korea and Japan, 10 May 2026 (link).

4 OPPORTUNITIES AND CHALLENGES FOR EUROPE

Expanding launch capacity is not an end in itself, without the ability to serve European payloads. Europe's sovereignty will continue to be challenged if upcoming military demand is not served by indigenous launch options **at will**.

While Ariane 6's return to flight is critical for Europe's access to space today, the expected volume and need for flexibility of governmental and commercial customers alike will be hard to satisfy with the currently planned cadence. With national military assets (including constellations) on the rise, the European Union's IRIS², an increased ESA budget, and aggressive commercial timelines to satisfy ITU deadlines drive expectations for aggressive launch cadence benchmarks. Plans to further increase the Ariane 6 launch cadence, targeting 12, 15 or 20 launches per year, are underway.[18] Despite clear demand signals, this expansion seems to be primarily driven by institutional rather than market forces.

Spaceports, launchers, and mobile platforms all need to develop in step with a framework that ties innovation, regulation, and procurement together into a coherent, responsive launch capability that serves Europe's strategic and sovereignty objectives in times of accelerated geopolitical tensions.

Most crucially, a credible responsive launch capability would also strengthen Europe's hand in international partnerships, reducing asymmetric dependency and enabling cooperation on more equal terms. International partners will also be essential to developing the full spectrum of European responsive space capabilities, with partners such as Canada, Japan, and Australia seemingly a natural avenue for cooperation. Underpinning this, Europe should develop a TSA-inspired technology safeguards framework to ensure sensitive technologies are handled responsibly on foreign territory.

As Europe is in the process of revamping its full suite of capabilities to access space, **responsiveness will become a defining measure of what sovereignty in space means**. In an environment where demand is surging for both scheduled launches and on-demand access to space, responsiveness will ultimately separate genuine autonomy from dependency.

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Point of Contact: Lina Pohl, ESPI Research Fellow seconded by DLR (lina.pohl@espi.eu)
Available online: www.espi.eu
Editor and publisher: European Space Policy Institute (ESPI)
Schwarzenbergplatz 16 TOP1, 1010 Vienna, Austria
Telephone: +43 1 718 11 18 – 0
Email: office@espi.eu
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[18] SpaceIntelReport (2026): ESA assesses what it would take to increase Ariane 6 cadence to 15 missions or more per year, before 2030, June 2026 (link).



ESPI

European Space
Policy Institute

Schwarzenbergplatz 16, 1010 Vienna, AT

office@espi.or.at

www.espi.or.at